

Tangents - the return

Say f and g are differentiable and we have parametric equations

$$x = f(t) \quad y = g(t)$$

then

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \quad \text{given } \frac{dx}{dt} \neq 0$$

Note, $\frac{dy}{dx}$ is a function of t .

The second derivative

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{\frac{d}{dt} \frac{dy}{dx}}{\frac{dx}{dt}}$$

with this information we can now find tangent lines and concavity

Ex:

$$\underline{x = t^2, y = t^3 - 3t}$$

a) Let's look at the point $(3, 0)$

we first need to find t

$$3 = t^2 \text{ so } \underline{t = \pm\sqrt{3}}$$

$$\frac{dy}{dx} = \frac{3t^2 - 3}{2t}$$

$$\left. \frac{dy}{dx} \right|_{t=-\sqrt{3}} = \frac{-3}{\sqrt{3}} = -\sqrt{3}$$

$$\left. \frac{dy}{dx} \right|_{t=\sqrt{3}} = \frac{3}{\sqrt{3}} = \sqrt{3}$$

the Point was $(3, 0)$ so

$$y = \sqrt{3}(x-3), \quad y = -\sqrt{3}(x-3)$$

are two tangent lines to this point

b) When is tangent line horizontal?

$$\frac{dy}{dx} = \frac{3t^2 - 3}{2t} = 0 \quad \text{when } 3t^2 - 3 = 0$$

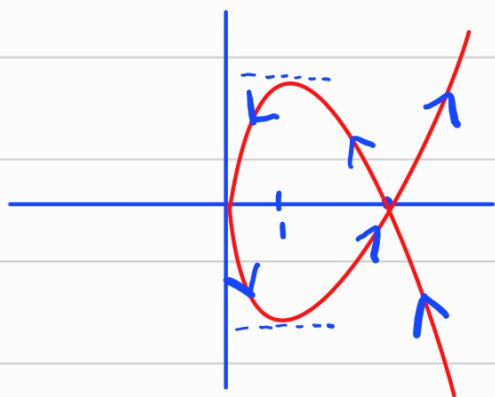
$$\boxed{t = \pm 1} \rightsquigarrow \begin{matrix} (1, 2) \\ (1, -2) \end{matrix}$$

c) When is Curve Concave up/down

$$\begin{aligned}\frac{d^2 y}{dx^2} &= \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}} = \frac{\frac{d}{dt} \left(\frac{3t^2 - 3}{2t} \right)}{2t} = \frac{\frac{d}{dt} \left(\frac{3}{2}t - \frac{3}{2t} \right)}{2t} = \frac{\frac{3}{2} + \frac{3}{2t^2}}{2t} \\ &= \frac{3}{4t} + \frac{3}{4t^3} = \frac{3t^2 + 3}{4t^3}\end{aligned}$$

$3t^2 + 3 > 0$ always, but $4t^3 > 0$ for $t > 0$
 < 0 for $t < 0$

So, Curve is concave up for $t > 0$,
and concave down for $t < 0$



Ex: A cycloid is an important curve found by following a single point on a circle as it rolls.



The curve is parameterized by

$$x = r(\theta - \sin \theta), \quad y = r(1 - \cos \theta)$$

we find $\frac{dy}{dx} = \frac{\sin \theta}{r(1 - \cos \theta)}$

- which has horizontal tangent lines when $\sin(\theta) = 0$
but $\cos(\theta) \neq 1$, i.e. when $\theta = \pi + 2k\pi \quad k \in \mathbb{Z}$
- when $\theta = 2k\pi$, there is a vertical tangent
as $1 - \cos \theta = 0$.

Area (Integration)

Usually, if $y=f(x)$, the area under the curve on $[a,b]$ is

$$A = \int_a^b f(x) dx = \int_a^b y dx$$

If we have a parameterized curve

$$x=f(t), \quad y=g(t)$$

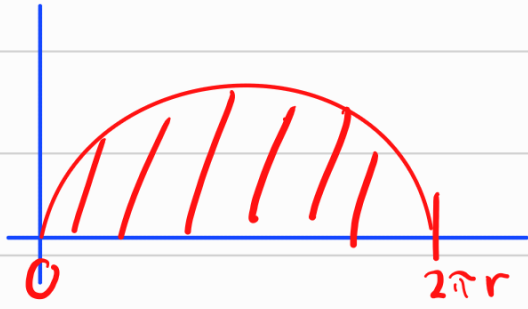
then

$$A = \int_a^b y dx = \int_a^b g(t) dx = \int_\alpha^\beta g(t) f'(t) dt$$

where α, β are the t values such that

$$x(\alpha) = a, \quad x(\beta) = b \quad \text{or vice versa}$$

Ex: Area under one arc of cycloid



$$x = r(t - \sin t)$$

$$y = r(1 - \cos t)$$

$$dx = r(1 - \cos t) dt$$

$$\int_0^{2\pi r} y dx = \int_0^{2\pi} r(1 - \cos t) r(1 - \cos t) dt$$

$$= r^2 \int_0^{2\pi} (1 - \cos t)^2 dt$$

$$= r^2 \int_0^{2\pi} 1 - 2\cos t + \cos^2 t dt$$

$$= r^2 \left[t \Big|_0^{2\pi} - 2\sin t \Big|_0^{2\pi} + \int_0^{2\pi} \cos^2 t dt \right]$$

$$= r^2 \left[2\pi + \frac{1}{2} \int_0^{2\pi} 1 + \cos(2t) dt \right]$$

$$= r^2 \left[2\pi + \frac{1}{2} t \Big|_0^{2\pi} + \sin(t) \Big|_0^{2\pi} \right]$$

$$= r^2 (2\pi + \pi + 0)$$

$$= 3\pi r^2$$

Arc Length

We will see that old ideas will have similar calculations in Parametric form.

For example, the old formula for arc length seen in Section 8.1 was

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

Using

$$dx = \frac{dx}{dt} dt \quad \text{and} \quad dy = \frac{dy}{dt} dt$$

$$L = \int_a^b \sqrt{1 + \left(\frac{dy/dt}{dx/dt}\right)^2} \frac{dx}{dt} dt \quad \rightarrow \quad dx = dx \frac{dt}{dt} = \frac{dx}{dt} dt$$

$$= \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 \left(1 + \frac{(dy/dt)^2}{(dx/dt)^2}\right)} dt$$

Note α, β are new bounds in terms of t .

$$= \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Ex: The Cycloid! Again

$$x = r(t - \sin t), y = r(1 - \cos t)$$

Arc length from $x=0$ to $x=2\pi r$ (one revolution)
 $t=0$ $t=2\pi$

$$dx = r(1 - \cos t)$$

$$dy = r \sin t$$

$$L = \int_0^{2\pi} \sqrt{(r - r \cos t)^2 + (r \sin t)^2} dt$$

$$= \int_0^{2\pi} \sqrt{r^2 - 2r^2 \cos t + r^2 \cos^2 t + r^2 \sin^2 t} dt$$

$$= \int_0^{2\pi} \sqrt{r^2 - 2r^2 \cos t + r^2} dt$$

$$= \int_0^{2\pi} \sqrt{2r^2 - 2r^2 \cos t} dt$$

$$= r \int_0^{2\pi} \sqrt{2(1 - \cos t)} dt$$

$$= r \int_0^{2\pi} \sqrt{4 \sin^2(t/2)} dt$$

$\hookrightarrow \sin^2(t) = \frac{1}{2}(1 - \cos(2t))$

$$= r \int_0^{2\pi} 2 \sin(t/2) dt$$

$$= 4r \left[-\cos\left(\frac{t}{2}\right) \right]_0^{2\pi}$$

$$= 4r \left(-(-1) - -(1) \right)$$

$\cos(\pi) = -1$ $\cos(0) = 1$

$$= 8r$$

Note: we can always parameter the function
 $y = f(x)$ with
 $x = t, y = f(t)$

So arc length here is

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

Since $\frac{dx}{dt} = 1$

Notice this was the original formula we started with.

Surface Area (optional)

The old formula for surface area was

$$S = 2\pi \int_a^b y \, ds \quad \text{where } ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

modifying this for parametric functions we get

$$S = 2\pi \int_a^b y \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Ex: Surface area of a sphere

A sphere is the rotation of a semicircle around X-axis. we parametrize this semicircle as

$$x = r \cos t \quad y = r \sin t \quad 0 \leq t \leq \pi$$

$$\frac{dx}{dt} = -r \sin t \quad \frac{dy}{dt} = r \cos t$$

$$S = 2\pi \int_0^\pi r \sin t \sqrt{(-r \sin t)^2 + (r \cos t)^2} dt$$

$$= 2\pi \int_0^{\pi} r \sin t \sqrt{r^2 \sin^2 t + r^2 \cos^2 t} dt$$

$$= 2\pi \int_0^{\pi} r^2 \sin t dt$$

$$= 2\pi r^2 [-\cos t]_0^{\pi}$$

$$= 2\pi r^2 (-(-1) - -(1))$$

$$= 4\pi r^2$$

Which matches the result in 8.2 but was a bit more straight forward.